YUKAI SONG

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SUMMARY

Ph.D. Candidate specializing in reliability-aware Machine Learning (ML) and Signal Processing for high-stakes healthcare systems, with a specialized focus in EEG-based modeling. Experienced in improving **model robustness, calibration, and failure-mode analysis** under noisy and real-world clinical conditions. Skilled in designing **deterministic, reproducible ML pipelines** and multi-stage decision frameworks for deployment-ready clinical algorithms.

RESEARCH EXPERIENCE

Mitigating High Confidence Errors in Sleep Stage Classification Using EEG Signals — **Dec 2025 – Present**

- Developed a **dual-expert (Transformer & Feature-based)** two-stage reliability pipeline for EEG, explicitly designed for **Algorithm Hardening** in clinical systems.
- Implemented a **risk-aware routing policy** based on **cross-model disagreement** (Type-1 label mismatch and Type-2 distributional divergence) to effectively detect and escalate high-risk cases.
- Executed **20-fold leave-one-subject-out (LOSO)** evaluation on Sleep-EDF-20, identifying **1,923 High-Confidence Errors (HCEs)**, a critical failure mode in clinical data.
- Achieved practical risk reduction: escalated **only 18.8% of samples to stage 2** while reducing the accepted-sample HCE rate to 1.62% (≈ **64.4% reduction**).

Two-Stage Voting for Robust and Efficient Suicide Risk Detection on Social Media — **Apr 2025 – Present**

- Designed a two-stage voting architecture with length-confidence routing using **a fine-tuned BERT classifier** to efficiently resolve high-confidence explicit cases, filtering **67.6%** of inputs and enabling cost-efficient reasoning under uncertainty.
- **Stage 2 resolves ambiguous cases** via two ensemble pathways: (i) **BERT + LLM Agents Voting** for maximum recall on subtle implicit ideation, and (ii) **BERT + ML** Voting over novel LLM-extracted psychological features for interpretability.
- Achieved strong cross-domain generalization, reaching up to **99.7% F1** while reducing the cross-domain performance gap to **<2%**.
- Implemented a scalable **PyTorch-based system** supporting this cascaded design and multi-model voting. Paper Link

Control-Aware Neural Network Pruning for Real-Time CPS Deployment — **Sep 2023 – Jul 2025**

- Designed a modular neural network pruning framework in PyTorch for **real-time, resource-constrained deployment.**
- Integrated **system-level and closed-loop safety constraints** into the pruning process to jointly optimize model **latency and end-to-end system performance.**
- Achieved up to **148×** acceleration while preserving system-level safety guarantees under closed-loop simulation.

Deep Learning for Contactless Human Motion Detection (RF-based) — **Sep 2021 – Dec 2022**

- **Performed Signal Processing Refinement** on RF-based human motion data, including applying a **Butterworth low-pass filter** for **artifact and noise handling**.
- Implemented and benchmarked multiple deep learning models in **PyTorch** (CNN, RNN, ResNet variants) to improve baseline classification accuracy from 93.8% to 98.6%.
- Proved the critical impact of **data preprocessing (PCA for dimension reduction)** over architectural changes, achieving a final accuracy of 99.1% with LSTM + PCA. Paper Link

SELECTED PUBLICATIONS

- **Song, Y.**, et al. Two-Stage Voting for Robust and Efficient Suicide Risk Detection on Social Media. Manuscript in preparation for submission to EMNLP 2026.
- **Song, Y.**, Taylor, W., Ge, Y., *et al.* Evaluation of deep learning models in contactless human motion detection system for next generation healthcare. *Sci Rep* 12, 21592 (2022).
- **Song, Y.**, Taylor, W., Ge, Y., *et al.* (2021). Design and Implementation of a Contactless AI-enabled Human Motion Detection System for Next-Generation Healthcare. In IEEE SmartIoT, pp. 112–119. **(Best Paper Award)**

TECHNICAL SKILLS

LLM & Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, OpenAI API, scikit-learn
Languages & Tools: Python, C/C++, CUDA, MATLAB, AWS, Docker, Git
Specialized Skills: Signal Processing, Safety/Alignment Metrics, Model Pruning/Compression, RAG, Agentic AI

EDUCATION

University of Pittsburgh — Ph.D. in Electrical & Computer Engineering (GPA: 4.0)
Swanson School of Engineering, Pittsburgh, PA | Aug 2023 – Present
Advisor: Prof. Jingtong Hu, Prof. Zhi-Hong Mao
Columbia University — M.S. in Electrical Engineering (GPA: 3.96)
The Fu Foundation School of Engineering and Applied Science, New York, NY | Sep 2021 – Feb 2023
University of Electronic Science and Technology of China (UESTC) — B.S. in Electronic Information Engineering
Glasgow College, Chengdu, China | Sep 2017 – Jun 2021
GPA: 3.92 (Rank: 1st / 257) | National Scholarship (Top 2%) | First Class Honours, Joint Program with University of Glasgow